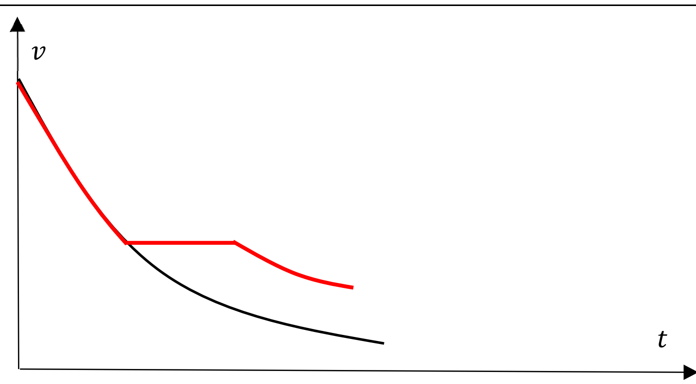
Question	Answer	Marks
7a	There is a magnetic flux of 1 **weber** through a surface if a magnetic field of flux density of 1T exists perpendicularly to an area of 1 m^2	**B1** **B1**
7bi	As AB moves from P towards Q, magnetic flux linkage over the area ABCD enclosed by the frame increases resulting in an induced e.m.f. generated in the frame By Lenz's Law, an induced current flows in the anticlockwise direction. This results in a magnetic force that acts on AB towards the left. Alternative: As AB enters the magnetic field, magnetic force acts on the electrons in AB driving them in a clockwise direction around the rectangular frame. The induced current flowing anti-clockwise in the frame results in a magnetic force that acts on AB towards the left.	**B1** **B1** **B1** **B1** **B1** **B1**
7bii	The magnetic flux (linkage) is given by $\Phi = BA = B(wx)$ where *x* is the distance AB has moved past P. Hence the induced emf is given by $E = \frac{d\Phi}{dt} = Bw\frac{dx}{dt} = Bwv$	**B1** **B1**
7biii	Induced current is thus $I = \frac{Bwv}{R}$ The braking force that acts on AB is thus $F = -BIw = -\frac{B^2w^2v}{R}$ And the acceleration given by $a = \frac{F}{m} = -\frac{B^2w^2v}{mR}$ OR applying *P* = *Fv*	**B1** **B1** **B1**
7ci	Distance 2*L* = Area under *v* – *t* graph = ½ (26 + 16 x 2 + 10 x 2 + 6 x 2 + 3.5)(5) = 233.75 m Range (211, 257) m PQ = *L* = 116.9 m	**B1** **B1** **A1**
7cii	The retarding force only acts on the train when there is a changing magnetic flux through the frame. There is no force on the train when the whole frame is within the magnetic field. Increasing the length PQ does not change the exit speed.	**M1** **A1**
7ciii	 Same slope [B1], plateau [B1], takes a shorter time to pass through the field, higher speed as it leaves the magnetic field [B1]	

7civ	To include ***multiple regions of magnetic field*** that can be activated individually when malfunction occurs. Use of electromagnet to generate the magnetic field so that the field can be strengthened by ***increasing the current in the electromagnet*** when malfunction occurs. Do not accept any modifications that relate to the dimensions of the train carriage as it is not possible to vary these during a malfunction.	**A1** **A1**
	Max Marks	**20**

Question	Answer	Marks
8(a)	Every point mass attracts every other point mass with a force that is directly	**A2**